

point error in the calculation of range that kept compounding over time. Deploying the software in a near-production like environment and stress testing it over longer-than-usual running times might have exposed the rounding off error before it had a chance to cause real harm.

An exhaustive automated testing suite as well as better deployment practices would have gone a long way to reduce the count of such errors. Another important facet of this is how non-functional needs (like logging floating point operation errors) though seemingly useless can sometimes be of immense help in identifying issues within the code base.

Case 5: NORAD reports U.S. was under missile attack (1980)

A third military goof up in the US (a bit of a pattern during that period) occurred in 1980 when a series of incorrect missile attack notifications were raised by the NORAD computers. What was even more shocking however is that this incident was reported to be “not a rare occurrence” and the defense alerts were usually verified with “human elements” of the system according to the then chief security adviser. Unfortunately the exact cause of the above case was never revealed, only that a single 46 cent chip was responsible for it. Experts have expressed skepticism regarding the validity of this explanation given another statement by the same security adviser where he said that the fault was “never reproduced during tests”. Without a concrete root cause, diagnosing and consequently attempting a fix for the problem would be futile, thus we are not going to attempt that on this case study.



Thank you for not starting WW3

Case 6: Man deletes entire company with single command (2016)

The latest of all the case studies, this story was the subject of much hype and speculation where a man had claimed to have accidentally deleted all data from his web hosting server including his client's. Later on though he posted back saying it was a joke but was based on an actual experience of a client's. This case is of specific significance here because if true, regardless of the source